

Artificial Intelligence

Course Information

- **Course Code:** CSC462 Course
- **Title:** Artificial Intelligence
- **Credit Hours:** 4 (3,1)

Text and Reference Books

- **Textbook:** Artificial Intelligence: A Modern Approach, Russell, S., and Norvig, P., Pearson, 2020.
- **Reference Book:** Artificial Intelligence Basics: A Non-Technical Introduction, Taulli, T., Apress, 2019.



Machine Learning

Zeenat

- My name is zeenat. I can learn everything automatically from experiences. Can you learn?"
- **Machine Learning (ML):**
- Yes, I can learn too! I learn from data. By analyzing patterns and trends in past experiences (data), I improve over time and make better decisions. While you learn naturally as a human, I use algorithms to process information and adaption.

What is Machine Learning?

- **Machine Learning (ML)** is a branch of **Artificial Intelligence (AI)** that allows computers to learn from data and improve their performance on a task **without being explicitly programmed**.
- In **normal programming**, we give the computer **exact instructions** on what to do.
- In **machine learning**, we **don't** write those exact rules.

Instead:

- We give the computer **data** (examples of input and output).
- The computer **learns the rules or patterns by itself** from the data.

For example:

You want a program to decide if an email is *spam* or *not spam*.

- You don't write rules like "if email contains the word 'lottery' then spam."
- You provide thousands of **emails labeled as spam or not spam**.

What is Machine Learning?

Example :

- If a system improves its performance **P** on a task **T** through experience **E**, then the system is said to **learn** from **E** with respect to **T** and **P**.

This is based on **Tom Mitchell's classic definition (1997):**

- *“A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E .”*

Machine learning vs. artificial intelligence



Definition



Artificial Intelligence (AI):
A broad field of computer science that aims to create systems that can **think, learn, and act intelligently**, like humans.



Machine Learning (ML): A **subset of AI** that focuses on systems that can **learn from data** and **improve automatically** without explicit programming.

Machine learning vs. artificial intelligence



2. Scope



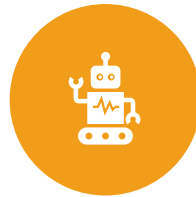
AI: The bigger concept; includes reasoning, learning, problem-solving, perception, and language understanding.



ML: A smaller part of AI focused only on **learning from data**.



3. Approach



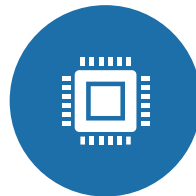
AI: Tries to make a machine **think and behave like humans**.



Can involve **rules, logic, and knowledge-based systems**.



ML: Tries to make a machine **learn from data**.



Always involves **data and learning algorithms**.

How Does Machine Learning Work?



Machine Learning works through the following steps:



a) Input Data



The process starts by collecting data, which could be anything—images, numbers, text, or sensor readings.



For example, if you want to teach a computer to differentiate between apples and oranges, you'll need a dataset of images of apples and oranges.



b) Training

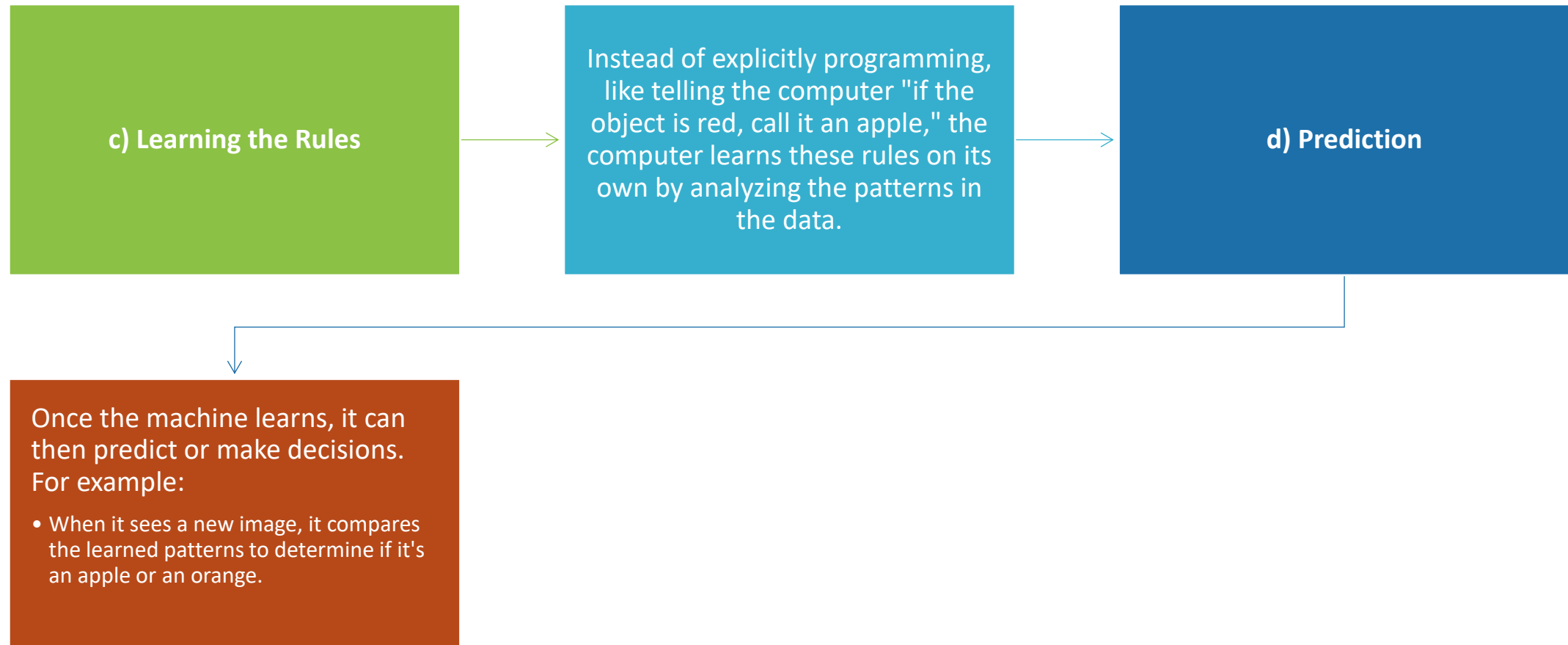


The collected data is used to "train" the machine. During training, the computer studies the patterns in the data. For example:

Apples might often appear red, round, and shiny.

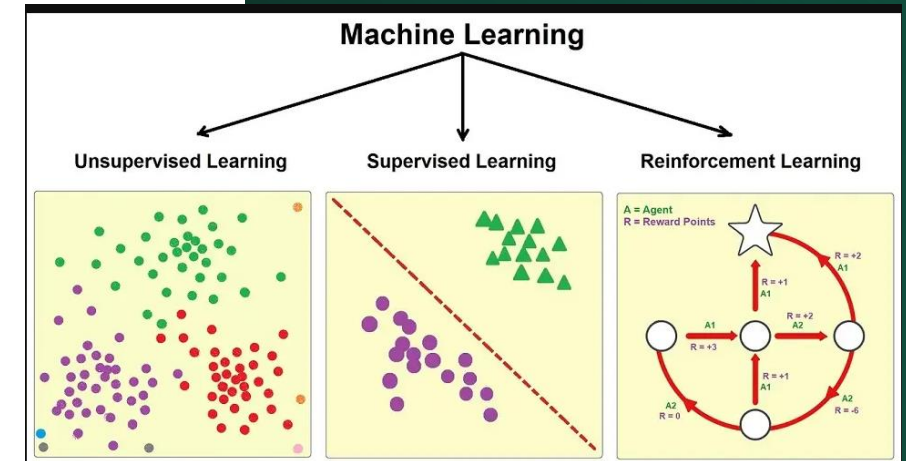
Oranges might often appear orange, round, and textured.

How Does Machine Learning Work?



Main Types of Machine Learning

- **Supervised Learning**
- **Unsupervised Learning**
- **Reinforcement Learning**



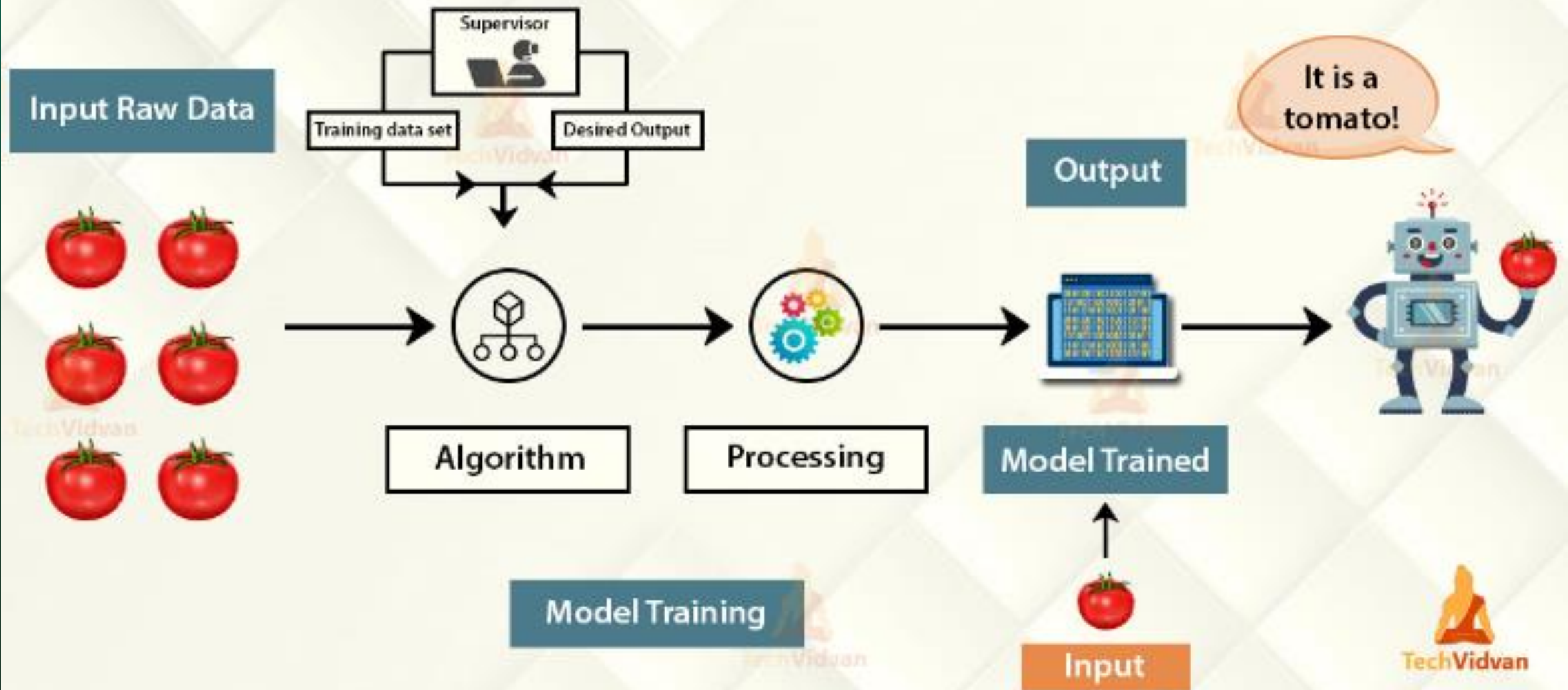
Supervised Learning

- Trains models on labeled data to predict or classify new, unseen data.
- Supervised Learning is based on **supervision**. It means in the supervised learning technique, we train the machines using the **"labelled" dataset**, and based on the training, the machine **predicts the output**. Here, the labelled data specifies that some of the **inputs are already mapped to the output**. More precisely, we can say; first, we train the **machine with the input and corresponding output**, and then we ask the machine to predict the **output using the test dataset**.

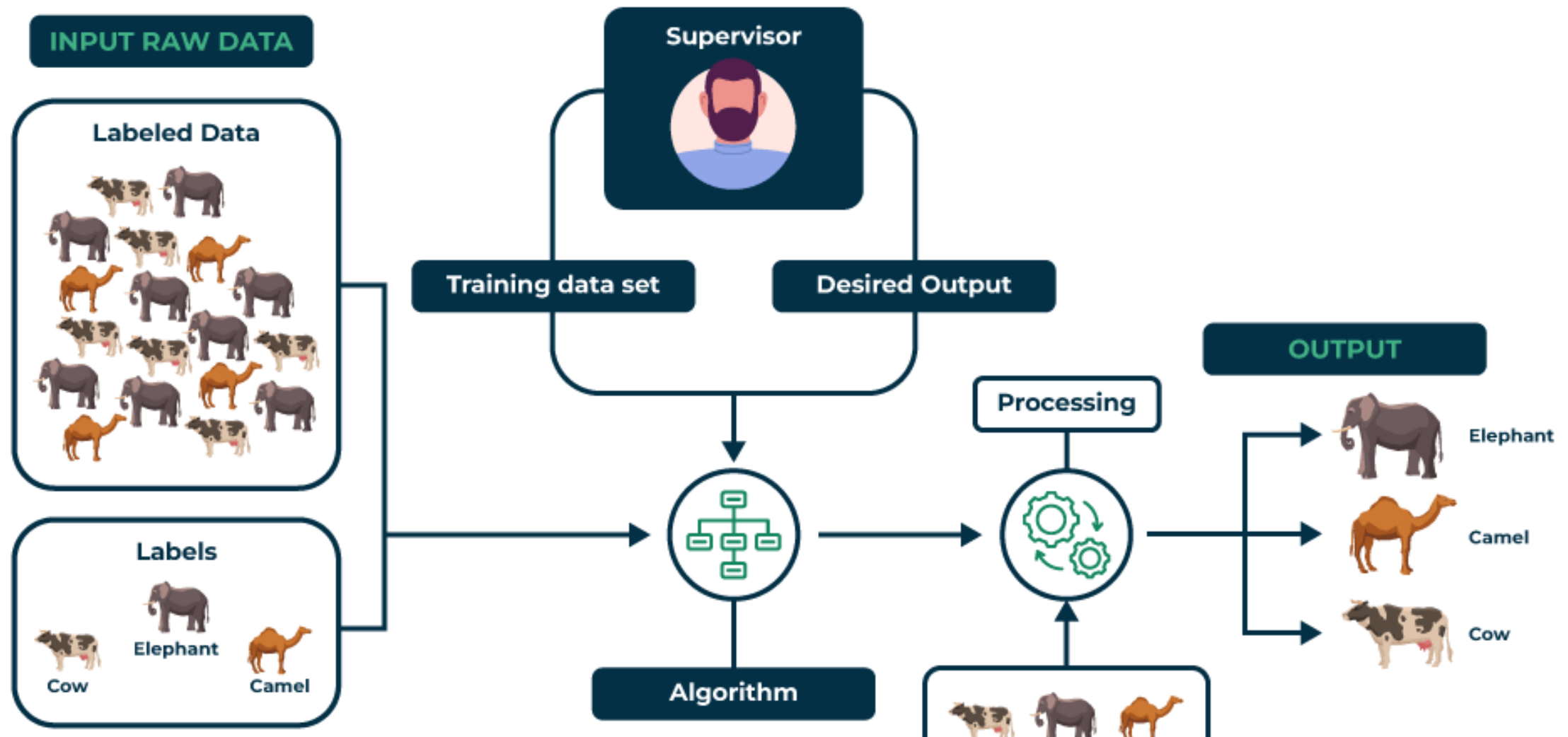
supervised learning with an example

- Suppose we have an input dataset of cats and dog images. So, first, we will provide the training to the machine to understand the images, such as the **shape & size of the tail of cat and dog, Shape of eyes, colour, height (dogs are taller, cats are smaller), etc.** After completion of training, we input the picture of a cat and ask the machine to identify the object and predict the output. Now, the machine is well trained, so it will check all the features of the object, such as height, shape, colour, eyes, ears, tail, etc., and find that it's a cat. So, it will put it in the Cat category. This is the process of how the machine identifies the objects in Supervised Learning.

Supervised Learning in ML



Supervised Learning



Categories of Supervised Machine Learning

- Supervised learning can be grouped further in two categories of algorithms:
- **Classification**
- **Regression**

Classification

- Classification algorithms are used to solve the classification problems in which the output variable is categorical, such as **"Yes" or No, Male or Female, Red or Blue, etc.** The classification algorithms predict the categories present in the dataset. Some real-world examples of classification algorithms are **Spam Detection, Email filtering, etc.**
- It is used to predict Discrete value

Regression

- Regression algorithms are used to predict continuous output variables, such as market trends, weather prediction, etc.

Difference between Regression and Classification

- In Regression, the output variable must be of continuous nature or real value.
- Regression Algorithms are used with continuous data.
- The regression Algorithm can be further divided into Linear and Non-linear Regression.
- In Classification, the output variable must be a discrete value.
- Classification Algorithms are used with discrete data.
- The Classification algorithms can be divided into Binary Classifier and Multi-class Classifier.

Scenario 1:

- A school wants to predict if a student will **pass or fail** based on their previous exam scores, attendance, and participation in class activities.
- **Question:** Is this classification or regression?

- **Answer: Classification**

Explanation: The outcome is categorical: "pass" or "fail." Classification problems deal with predicting discrete labels.

Scenario 2:

- A real estate agent needs to estimate the **price** of a house based on its size, location, number of bedrooms, and age.

- **Answer: Regression**

Explanation: The outcome is a continuous value (the price of the house). Regression problems deal with predicting numerical values.

Scenario 3:

- A hospital wants to predict whether a patient has **diabetes** or not, using their medical history and test results.

- **Answer: Classification**

Explanation: The outcome is categorical: either "has diabetes" or "does not have diabetes."

Scenario 4:

- A company wants to predict the **number of products sold** next month based on last year's sales data.

- **Answer: Regression**

Explanation: The outcome is a continuous numerical value (number of products sold).

Scenario 5:

- A sports team is trying to classify whether a match will be a **win, loss, or draw**, based on team performance metrics.

- **Answer: Classification**

Explanation: The outcome is categorical with three classes: "win," "loss," or "draw."

- You are working as a data scientist for a ride-sharing company like Uber. The company wants you to solve several business problems using data. Here are the tasks you are assigned:
 1. Predict whether a **customer will cancel their ride** or not based on factors like the distance to the pickup location, time of booking, and weather conditions.
 2. Estimate the **total revenue** the company will earn in the next month based on historical data of completed rides.
 3. Predict the **type of vehicle** a customer is most likely to book (e.g., Economy, SUV, Luxury) based on their previous bookings, travel distance, and preferences.
 4. Forecast the **average ride duration** in minutes for the upcoming week based on the day of the week, time of day, and traffic conditions.
 5. Determine whether a particular **driver's rating is above or below 4 stars** based on customer reviews and feedback.

Question: For each task, identify whether it is a **classification** or **regression** problem.

- **Answers:**

- 1. Predict whether a customer will cancel their ride or not**

Answer: Classification

Explanation: The outcome is categorical: "cancel" or "not cancel."

- 2. Estimate the total revenue the company will earn in the next month**

Answer: Regression

Explanation: The outcome is a continuous numerical value (total revenue).

- 3. Predict the type of vehicle a customer is most likely to book**

Answer: Classification

Explanation: The outcome is categorical with multiple classes (Economy, SUV, Luxury).

- 4. Forecast the average ride duration in minutes for the upcoming week**

Answer: Regression

Explanation: The outcome is a continuous numerical value (ride duration in minutes).

- 5. Determine whether a particular driver's rating is above or below 4 stars**

Answer: Classification

Explanation: The outcome is binary: "above 4 stars" or "below 4 stars."

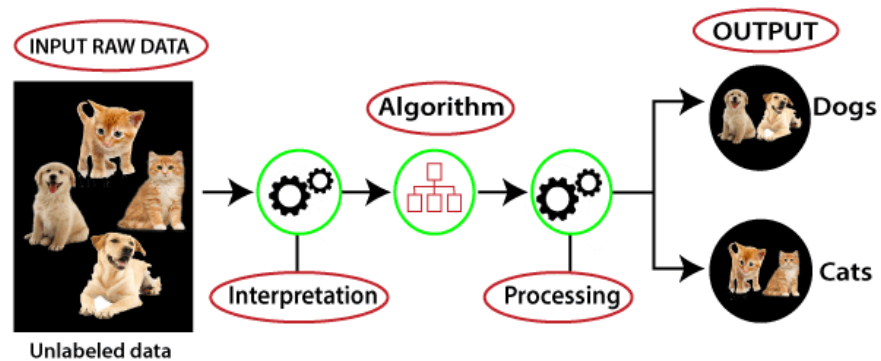
Unsupervised Learning

- Unsupervised learning is a learning method in which a machine learns without any **supervision**.
- The training is provided to the machine with the set of data that has **not been labeled**, classified, or categorized, and the algorithm needs to act on that **data without any supervision**

Example

- Suppose the unsupervised learning algorithm is given an input dataset containing images of different types of cats and dogs. The algorithm is never trained upon the given dataset, which means it does not have any idea about the features of the dataset. The task of the unsupervised learning algorithm is to identify the image features on their own. Unsupervised learning algorithm will perform this task by clustering the image dataset into the groups according to similarities between images.

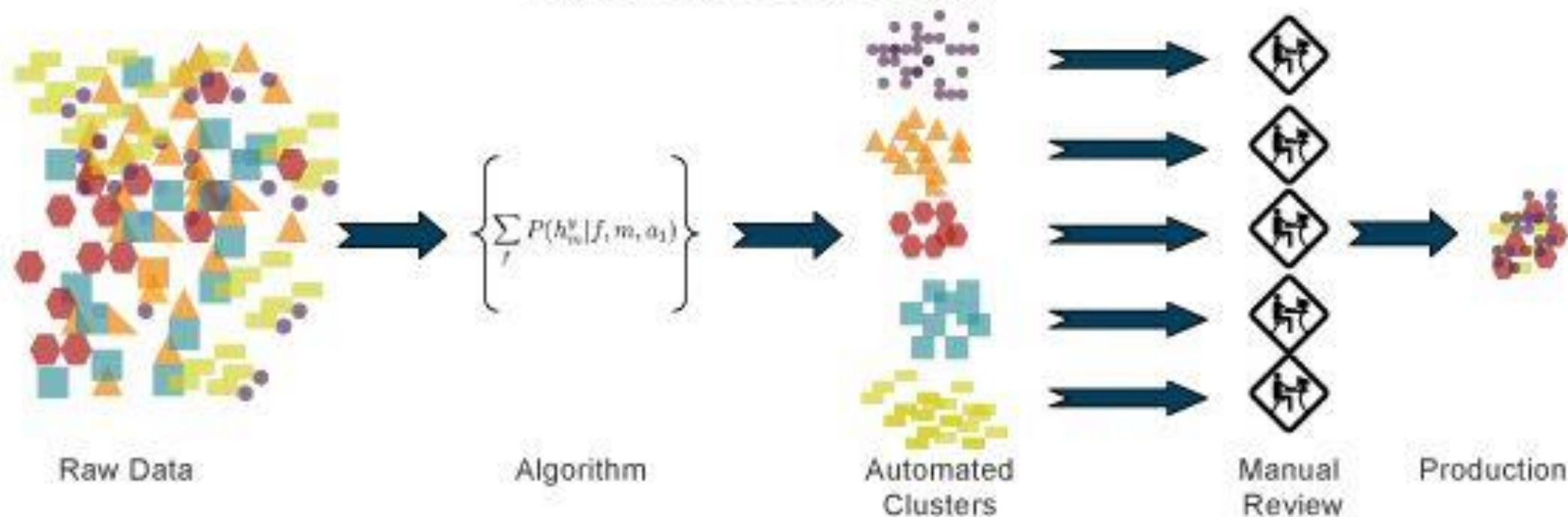
- Here, we have taken an **unlabeled input data**, which means **it is not categorized**, and **corresponding outputs are also not given**.



- Now, this unlabeled input data is **fed** to the **machine learning model** in order to train it. Firstly, it will **interpret the raw data** to find the **hidden patterns from the data** and then will apply suitable algorithms such as k-means clustering, Decision tree, etc.

UNSUPERVISED LEARNING

High reliance on algorithm for raw data, large expenditure on manual review for review for relevance and coding



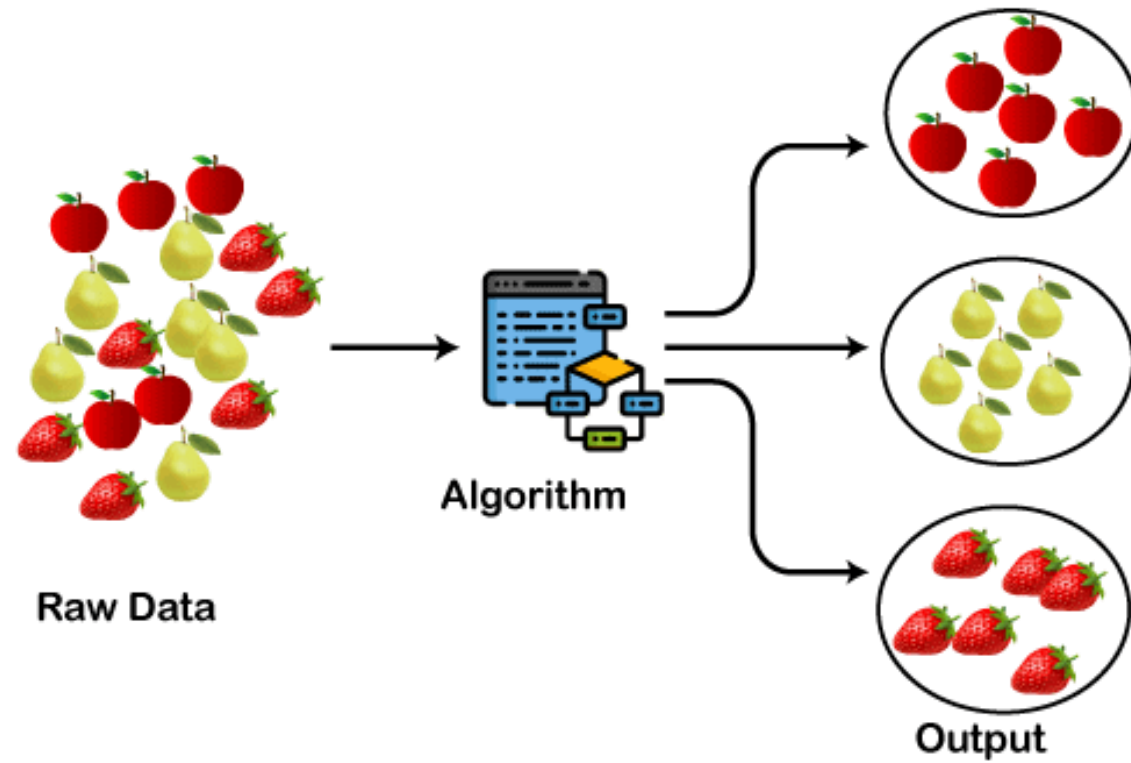
Types of Unsupervised Learning

- It can be further classified into two categories of algorithms:
- **Clustering**
- **Association**

Clustering

- Clustering is an **unsupervised machine learning technique** used to group similar data points into clusters based on their features or characteristics. The goal is to ensure that data points in the same cluster are **more similar to each other** than to those in other clusters.
- Unlike supervised learning, clustering does **not require labeled data** , it automatically discovers hidden patterns or structures within the dataset.

Clustering



Classification Models

Traditional (Classical) Models

Logistic Regression

K-Nearest Neighbors (KNN)

Decision Tree Classifier

Random Forest Classifier

Support Vector Machine (SVM)

Naïve Bayes

Gradient Boosting Classifier

AdaBoost Classifier

XGBoost Classifier

LightGBM Classifier

Regression Models

Linear Regression

Decision Tree Regressor

Random Forest Regressor

Support Vector Regressor (SVR)

K-Means Clustering

Supervised and Unsupervised Learning

Feature	Supervised Learning	Unsupervised Learning
Definition	The model learns from labeled data (data with known outputs).	The model learns from unlabeled data (data without known outputs).
Goal	To predict outcomes or classify new data based on past examples.	To find hidden patterns, structures, or relationships in data.
Data Type	Input data + corresponding output labels.	Only input data (no output labels).
Example of Data	Email + label (spam / not spam)	Customer purchase data (no labels)
Common Algorithms	Linear Regression, Logistic Regression, Decision Tree, Random Forest, SVM, KNN, Neural Networks	K-Means Clustering, Hierarchical Clustering, DBSCAN, PCA (Principal Component Analysis), Autoencoders
Output Type	Predicts continuous values (regression) or categories (classification).	Groups data into clusters or reduces dimensions .
Human Involvement	Requires labeled data prepared by humans.	No labeling needed — model explores patterns by itself.
Example Applications	Email spam detection, disease prediction, sentiment analysis.	Customer segmentation, market basket analysis, anomaly detection.
Evaluation	Measured using accuracy, precision, recall, F1-score, RMSE, etc.	Harder to evaluate; uses metrics like silhouette score or visualization.

Supervised Machine Learning with KNN

- **K-Nearest Neighbors (KNN)** is a **supervised learning algorithm** used for both **classification** and **regression**, but it is most commonly used for **classification** tasks.
- It is a **lazy learning** algorithm meaning it does not build a model during training. Instead, it **stores the training data** and makes predictions only when new data needs to be classified.

Working of KNN Algorithm

- The KNN algorithm works based on the idea that **similar data points exist close to each other** in feature space.
- **Steps:**
- Choose the number of neighbors (**K**).
- Calculate the **distance** (usually Euclidean distance) between the new data point and all training points.
- Select the **K nearest neighbors** to the new point.
- For **classification**: The class with the majority among the K neighbors is assigned to the new data point.
For **regression**: The average value of the K neighbors is used as the prediction.

Distance Metric (Commonly Used)

- For two points **A**(x_1, y_1) and **B**(x_2, y_2), the **Euclidean distance** is:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Working of KNN

Movie Imdb

KNN

Classification (predicting classes)

e.g Predicting movie Genre

IMDb Rating	Duration	Genre
8.0 (Mission Impossible)	160	Action
6.2 (Predator 2)	170	Action
7.2 (Rocky and Bullwinkle)	168	Comedy
8.2 (OMG 2)	155	Comedy

Now predict the genre of "Barbie" movie with IMDb rating 7.4 and duration 114

Step 1: Calculation Distance (Euclidean distance)

Calculate the Euclidean distance b/w the new movie and each movie in the dataset

$$d(p, q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\begin{aligned} x_1 &= 7.4 \\ y_1 &= 114 \end{aligned}$$

Working of KNN

$$\begin{aligned}\text{Distance to } (8.0, 160) &= \sqrt{(\overset{x_2}{7.4} - \overset{y_2}{8.0})^2 + (114 - 160)^2} = \approx 46.00 \\ \text{Distance to } (6.2, 160) &= \sqrt{(7.4 - 6.2)^2 + (114 - 170)^2} = \approx 56.01 \\ \text{Distance to } (7.2, 168) &= \sqrt{(7.4 - 7.2)^2 + (114 - 168)^2} = \approx 54.00 \\ \text{Distance to } (8.2, 155) &= \sqrt{(7.4 - 8.2)^2 + (114 - 155)^2} = \approx \textcircled{41.00}\end{aligned}$$

Step 2:

Select k Nearest Neighbors

Suppose $k=1$ (Neighbors)

Small distance 41.00

Best value $\textcircled{5}$

Step 3: Majority voting $\overset{k=3}{(A, C, C)}$
classification

Barbie move "C"

Machine learning libraries:

- **Pandas** - pandas is a fast, powerful, exible and easy to use open-source data analysis and manipulation tool, built on top of the Python programming language.
- **NumPy** NumPy is the fundamental package for array computing with Python.
- **Matplotlib** - Matplotlib is the plotting library for Python that provides an object-oriented API for embedding plots into applications.

Machine learning libraries:

- **Seaborn** - Seaborn is a Python data visualization library based on matplotlib. It provides a high-level interface for drawing attractive and informative statistical graphics.
- **Plotly** - The plotly Python library is an interactive, open-source plotting library that supports over 40 unique chart types covering a wide range of statistical, financial, geographic, scientific, and 3-dimensional use-cases.

Machine learning libraries:

- **Scikit-learn (sklearn)**
- **Scikit-learn** is one of the most popular and powerful **machine learning libraries in Python**. It provides simple and efficient tools for **data analysis, preprocessing, and model building**.
- It is built on top of:
 - **NumPy** – for numerical operations
 - **SciPy** – for scientific computations
 - **Matplotlib** – for visualization

Machine learning libraries:

- **Advantages of Scikit-learn**
- Easy to use and well-documented
- Covers almost all basic machine learning algorithms
- Works well with NumPy and Pandas
- **Scikit-learn** is the official name.
- **sklearn** is the module you import in Python.
- **It's a complete toolkit for building, training, and evaluating machine learning models.**

Supervised Machine Learning with KNN

Implement the k-Nearest Neighbors (KNN) algorithm for supervised learning.



Use Scikit-learn to implement the KNN algorithm for classification tasks. Evaluate the performance of the KNN model on different datasets.

```
1  # Import libraries
2  from sklearn.datasets import load_iris
3  from sklearn.model_selection import train_test_split
4  from sklearn.neighbors import KNeighborsClassifier
5  from sklearn.metrics import accuracy_score
6
```

- `load_iris`: Loads the famous **Iris dataset** from scikit-learn.
- `train_test_split`: Splits the dataset into **training** and **testing** subsets.
- `KNeighborsClassifier`: This is the **KNN algorithm** used for classification.
- `accuracy_score`: Calculates how accurate the model's predictions are.

Use Scikit-learn to implement the KNN algorithm for classification tasks.

```
|  
# Step 1: Load dataset  
data = load_iris()           # Load the Iris dataset  
X = data.data                # Features (inputs)  
y = data.target              # Labels (outputs)
```

- `load_iris()` loads the Iris dataset, which contains measurements of flowers.
- `data.data` → The **input features** (like sepal length, sepal width, etc.).
- `data.target` → The **target labels** (flower species: setosa, versicolor, virginica)

Use Scikit-learn to implement the KNN algorithm for classification tasks.

```
# Step 2: Split data into training and testing sets (80% train, 20% test)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

- **X** → Your **input features (independent variables)**

Example: sepal length, sepal width, etc.

- **y** → Your **labels or target (dependent variable)**

Example: flower species (setosa, versicolor, virginica).

- **test_size=0.2** → This means **20%** of the data will go into the **testing set**, and **80%** will go into the **training set**.

- You can change this to other ratios, e.g. 0.3 (30% test) or 0.1 (10% test).

- **random_state=42** → A **random seed number** that ensures the split is **always the same** each time you run the code.

- Without this, the data would be split differently every time (results change).

- Using a fixed number (like 42) makes the results **reproducible**

Use Scikit-learn to implement the KNN algorithm for classification tasks.

Use Scikit-learn to implement the KNN algorithm for classification tasks.

```
# Step 3: Create KNN model with k=3  
model = KNeighborsClassifier(n_neighbors=3)
```

- KNeighborsClassifier creates a KNN model.
- n_neighbors=3 means the algorithm looks at the **3 nearest neighbors** to decide the class of a data point.
- KNN is a **non-parametric** algorithm — it doesn't "learn" parameters, it just memorizes data.

Use Scikit-learn to implement the KNN algorithm for classification tasks.

- `fit()` trains the model using the **training dataset**.
- In KNN, this means it **stores** all training examples so that it can later compare new inputs with them.

```
# Step 4: Train the model  
model.fit(X_train, y_train)
```

Use Scikit-learn to implement the KNN algorithm for classification tasks.

```
# Step 5: Predict on both training and test data
y_train_pred = model.predict(X_train) # Predictions for training data
y_test_pred = model.predict(X_test)   # Predictions for test data
```


Use Scikit-learn to implement the KNN algorithm for classification tasks.

```
# Step 6: Evaluate accuracy for both
train_accuracy = accuracy_score(y_train, y_train_pred)
test_accuracy = accuracy_score(y_test, y_test_pred)

# Step 7: Print the results
print("Training Accuracy:", train_accuracy)
print("Testing Accuracy:", test_accuracy)
```

Difference between classification and clustering

Feature	Classification	Clustering
Type of Learning	Supervised Learning – Uses labeled data (data with known categories).	Unsupervised Learning – Uses unlabeled data (no predefined categories).
Goal	To assign new data points to known categories or classes .	To group data points into similar clusters based on patterns or similarity.
Example	Classifying emails as <i>spam</i> or <i>not spam</i> .	Grouping customers based on their buying behavior.
Training Data	Requires predefined labels during training.	No labels are provided; the algorithm finds structure on its own.
Output	Predicts a class label for each data point.	Produces clusters (groups) of similar data points.
Common Algorithms	Decision Tree, Random Forest, SVM, KNN, Logistic Regression.	K-Means, Hierarchical Clustering, DBSCAN.